HDOC Day_53 Report-Programming in C

Problem

Write a C program that displays the following menu and uses a switch statement to print the “Decreasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

Note: For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.

1. Decreasing right-angled triangle

Input the number of rows, n, and print a decreasing right-angled triangle using asterisks.

For n = 4:

* * * *

* * *

* *

*

2. Decreasing natural-number pattern

Input the number of rows, n. In the first row, print the natural numbers from 1 to n. Reduce the number of values printed by one in each successive row.

For n = 4:

1 2 3 4

1 2 3

1 2

1

3. Decreasing repeated-number pattern

Input the number of rows, n. Begin with n in the first row and decrease the value by one in each successive row. Print each value as many times as its value.

For n = 4:

4 4 4 4

3 3 3

2 2

1 4 3 2 1

3 2 1

21 Algorithm **Error! Bookmark not defined.**

4. Reverse-number triangle

Input the number of rows, n. In each row, print natural numbers in descending order, beginning with the value corresponding to the number of elements in that row and ending with 1.

For n = 4:

1. **Start** the program.
2. Declare an integer variable ch to store the user's choice.
3. Display the menu with the following options:
    - Choice 1: Print a decreasing right-angled triangle using asterisks. ○ Choice 2: Print natural numbers from 1 to n, decreasing the number of elements in each row. ○ Choice 3: Print each row's number repeatedly according to its value, starting from n and decreasing to 1.
    - Choice 4: Print natural numbers in descending order in each row.
4. Ask the user to **enter their choice**.
5. Read the value of ch.
6. Use a **switch statement** to process the choice:
    - If ch = 1, call decreasing_rat(). ○ If ch = 2, call decreasing_nnum(). ○ If ch = 3, call decreasing_rnum(). ○ If ch = 4, call decreasing_revnum().
    - For any other value, display **"invalid input"**.
7. In decreasing_rat():
    - Read the number of rows n.
    - Start from row n and decrease the number of stars by one in every row.

- Print the required stars.

8. In decreasing_nnum():
   - Read n. ○ Start with n elements in the first row. ○ Print numbers from 1 to the current row number.
   - Decrease the number of elements by one for each successive row.
9. In decreasing_rnum(): ○ Read n. ○ Start with value n. ○ Print the value i, i times.
   - Decrease i by one after every row until 1.
10. In decreasing_revnum(): ○ Read n. ○ Start with i = n. ○ Print numbers from i down to 1.
    - Decrease i by one for every successive row.
11. Return to main() after the selected function finishes.
12. End the program using return 0.

C Program Code

```
#include <stdio.h> void
decreasing_rat()
{   int
n;
  printf("enter no of rows\n");
scanf("%d",&n);          for(int
i=n;i>=1;i--)
  {
    for(int j=i;j>=1;j--)
    {
      printf("* ");
    }
    printf("\n");
  }
```

```
}
void decreasing_nnum()
{    int
n;
   printf("enter no of rows\n");
scanf("%d",&n);     for(int
i=n;i>=1;i--)
   {
      for(int j=1;j<=i;j++)
      {
         printf("%d ",j);
      }
      printf("\n");
   }
}
void decreasing_rnum()
{    int
n;
   printf("enter no of rows\n");
scanf("%d",&n);             for(int
i=n;i>=1;i--)
   {
      for(int j=i;j>=1;j--)
      {
         printf("%d ",i);
      }
      printf("\n");
   }
```

```
}
void decreasing_revnum()
{    int
n;
   printf("enter no of rows\n");
scanf("%d",&n);              for(int
i=n;i>=1;i--)
   {
      for(int j=i;j>=1;j--)
      {
         printf("%d ",j);
      }
      printf("\n");
   }
}
int main()
{    int
ch;
   printf("choice 1 is to print a decreasing right-angled triangle using asterisks.\n");
printf("choice 2 is to  print the natural numbers from 1 to n. Reduce the number of values printed by one in each successive row.\n");     printf("choice 3 is to Begin with n in the first row and decrease the value by one in each successive row. Print each value as many times as its value.\n");     printf("choice 4 is to , print natural numbers in descending order, beginning with the value corresponding to the number of elements in that row and ending with 1.\n");     printf("enter your choice\n");     scanf("%d",&ch);     switch (ch)
   {
      case 1:
      {
```

```
            decreasing_rat();
break;
        }
        case 2:
        {
            decreasing_nnum();
            break;
        }
        case 3:
        {
            decreasing_rnum();
            break;
        }
        case 4:
        {
            decreasing_revnum();
            break;
        }
        default:
        {
            printf("invalid input\n");
break;
        }
    }
    return 0;
}
```

Test Cases

| TC02 | 2 | 4 | ```text

1 2 3 4

1 2 3

1 2

1

``` |

| TC03 | 3 | 4 | ```text

4 4 4 4

3 3 3

2 2

1

``` |

| TC04 | 4 | 4 | ```text

4 3 2 1

3 2 1

2 1

1

``` |

| TC05 | 1 | 3 | ```text

* * *

* *

*

``` |

| TC06 | 2 | 3 | ```text

1 2 3

1 2

1

``` |
```

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| TC07 | 3 | 3 | ```text

3 3 3

2 2

1

``` |

| TC08 | 4 | 3 | ```text

3 2 1

2 1

1

``` | | TC09 | 5 | — |

```text invalid input ```

|
```